
BIM488 Introduction to Pattern Recognition

Face Recognition

Outline

- Introduction
- Applications
- Facial features
- Face recognition approaches
- Eigenface method

Introduction

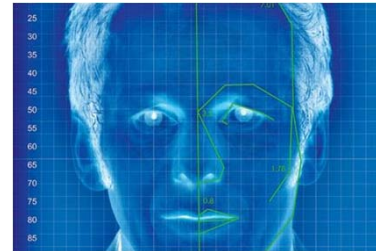
- Face recognition is a specific field of image recognition.
- The task is to enable a machine to identify or verify a face from a digital image or video.

Applications

- Criminal identification
- Security systems
- Image and film processing
- Human-computer interaction
- etc.

Facial features

- Every face has numerous, distinguishable **landmarks**, the different peaks and valleys that make up facial features such as:
 - Distance between the eyes
 - Width of the nose
 - Depth of the eye sockets
 - The shape of the cheekbones
 - The length of the jaw line

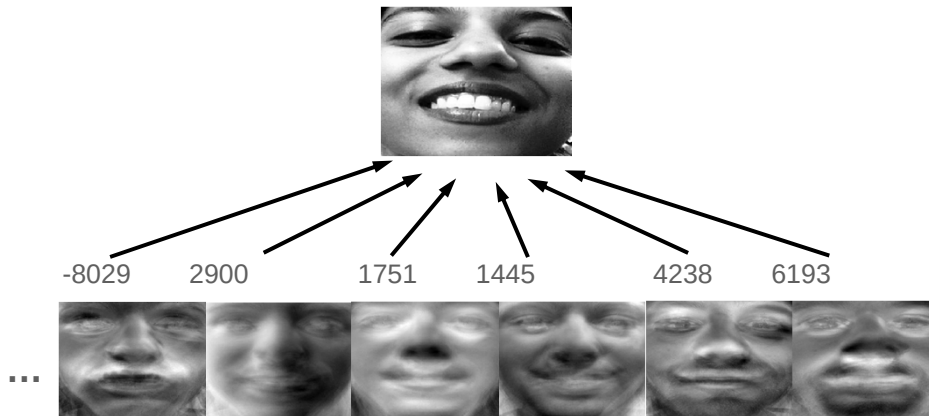


Face Recognition Approaches

- There are two fundamental approaches to the face recognition problem:
 1. Geometric (feature based), which looks at distinguishing features.
 2. Photometric (view based), which is a statistical approach that distill an image into values and comparing the values with templates to eliminate variances.
- As researcher interest in the subject continued, many different algorithms were developed.
- In this course, we will focus on Eigenfaces, which is one of the most popular methods in face recognition.

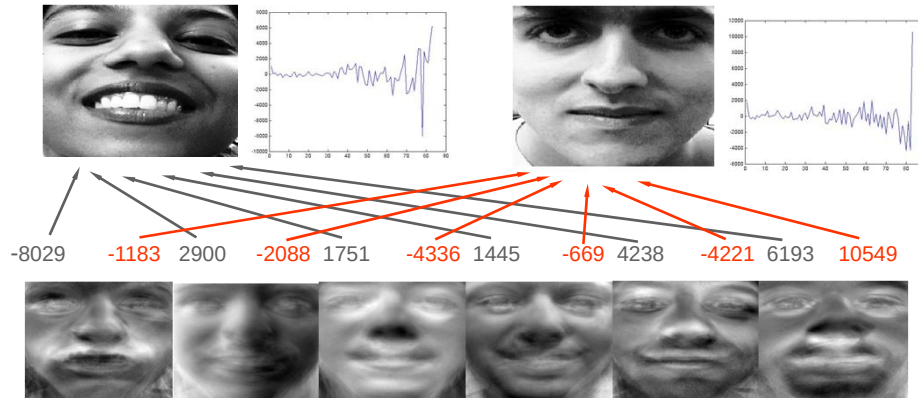
Eigenfaces: the idea

- Think of a face as being a weighted combination of some “component” or “basis” faces
- These basis faces are called eigenfaces



Eigenfaces: representing faces

- These basis faces can be differently weighted to represent any face
- So we can use different vectors of weights to represent different faces



Learning Eigenfaces

Q: How do we pick the set of basis faces?

A: We take a set of real training faces



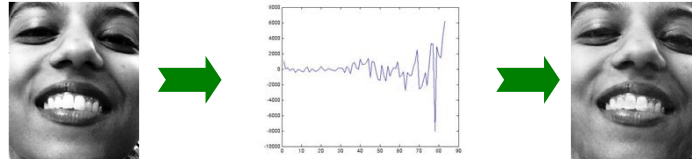
Then we find (learn) a set of basis faces which best represent the differences between them

We'll use a statistical criterion for measuring this notion of "best representation of the differences between the training faces"

We can then store each face as a set of weights for those basis faces

Using Eigenfaces: recognition & reconstruction

- We can use the eigenfaces in two ways
 1. We can store and then reconstruct a face from a set of weights



2. We can recognise a new picture of a familiar face

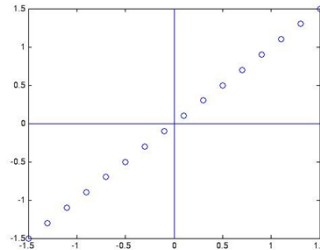


Learning Eigenfaces

- How do we learn them?
- We use a method called Principle Components Analysis (PCA)
- To understand this we will need to understand
 - What an eigenvector is
 - What covariance is
- But first we will look at what is happening in PCA qualitatively

Subspaces

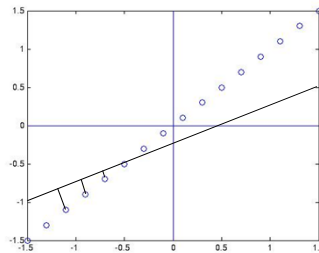
- Imagine that our face is simply a (high dimensional) vector of pixels
- We can think more easily about 2d vectors



- Here we have data in two dimensions
- But we only really need one dimension to represent it

Finding Subspaces

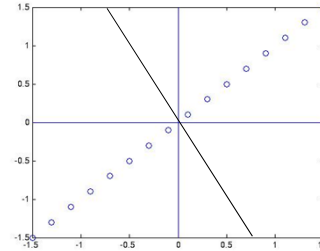
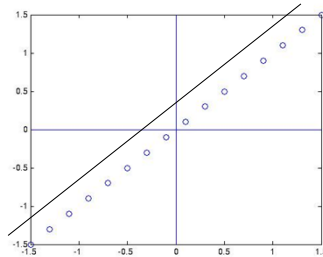
- Suppose we take a line through the space



- And then take the projection of each point onto that line
- This could represent our data in “one” dimension

Finding Subspaces

- Some lines will represent the data in this way well, some badly

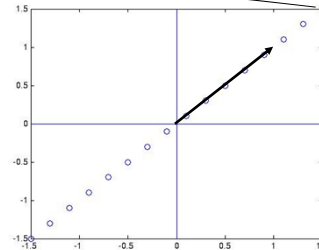


- This is because the projection onto some lines separates the data well, and the projection onto some lines separates it badly

Finding Subspaces

- Rather than a line we can perform roughly the same trick with a vector

$$\Phi_{16} = \frac{3}{2} \nu$$



$$\nu = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\Phi_i = \mu \nu$$

- Now we have to scale the vector to obtain any point on the line

Eigenvectors

- An eigenvector is a vector v that obeys the following rule:

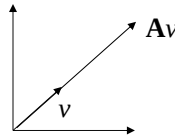
$$Av = \mu v$$

Where A is a matrix, μ is a scalar (called the eigenvalue)

e.g. $A = \begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix}$ one eigenvector of A is $v = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$ since

$$\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 12 \\ 8 \end{bmatrix} = 4 \times \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

so for this eigenvector of this matrix the eigenvalue is 4

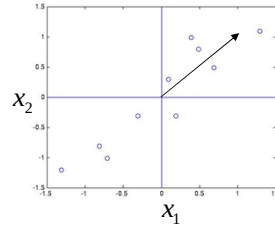


Eigenvectors

- We can think of matrices as performing transformations on vectors (e.g rotations, reflections)
- We can think of the eigenvectors of a matrix as being special vectors (for that matrix) that are scaled by that matrix
- Different matrices have different eigenvectors
- Only square matrices have eigenvectors
- Not all square matrices have eigenvectors
- An n by n matrix has at most n distinct eigenvectors
- All the distinct eigenvectors of a matrix are orthogonal (ie perpendicular)

Covariance

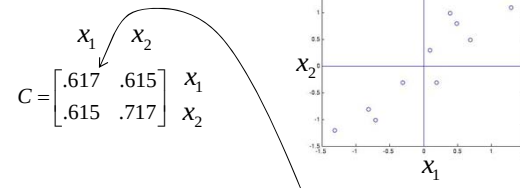
- Which single vector can be used to separate these points as much as possible?



- This vector turns out to be a vector expressing the direction of the correlation
- Here I have two variables x_1 and x_2
- They co-vary (y tends to change in roughly the same direction as x)

Covariance

- The covariances can be expressed as a matrix



- The diagonal elements are the variances e.g. $\text{Var}(x_1)$
- The covariance of two variables is:

$$\text{cov}(x_1, x_2) = \frac{\sum_{i=1}^n (x_1^i - \bar{x}_1)(x_2^i - \bar{x}_2)}{n-1}$$

Eigenvectors of the covariance matrix

- The covariance matrix has eigenvectors

covariance matrix

$$C = \begin{bmatrix} .617 & .615 \\ .615 & .717 \end{bmatrix}$$

eigenvectors

$$v_1 = \begin{bmatrix} -.735 \\ .678 \end{bmatrix}$$

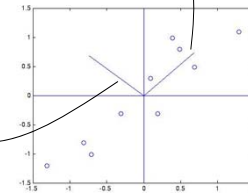
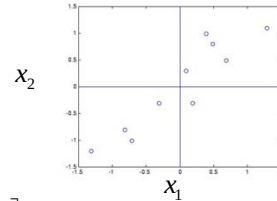
$$v_2 = \begin{bmatrix} .678 \\ .735 \end{bmatrix}$$

eigenvalues

$$\mu_1 = 0.049$$

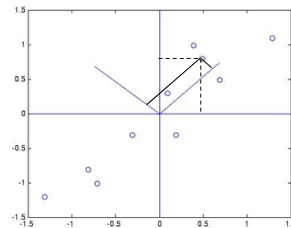
$$\mu_2 = 1.284$$

- Eigenvectors with larger eigenvalues correspond to directions in which the data varies more
- Finding the eigenvectors and eigenvalues of the covariance matrix for a set of data is termed principle components analysis



Expressing points using eigenvectors

- Suppose you think of your eigenvectors as specifying a new vector space



- i.e. I can reference any point in terms of those eigenvectors
- A point's position in this new coordinate system is what we earlier referred to as its "weight vector"
- For many data sets you can cope with fewer dimensions in the new space than in the old space

Eigenfaces

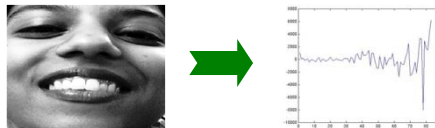
- All we are doing in the face case is treating the face as a point in a high-dimensional space, and then treating the training set of face pictures as our set of points
- To train:
 - We calculate the covariance matrix of the faces, or perform singular value decomposition (SVD).
 - We then find the eigenvectors and eigenvalues of that covariance matrix
- These eigenvectors are the eigenfaces or basis faces
- Eigenfaces with bigger eigenvalues will explain more of the variation in the set of faces, i.e. will be more distinguishing

Eigenfaces: image space to face space

- When we see an image of a face we can transform it to face space

$$\mathbf{w}_k = \mathbf{x}^i \cdot \mathbf{v}_k$$

- There are $k=1 \dots n$ eigenfaces $\mathbf{v}_k$
- The i^{th} face in image space is a vector $\mathbf{x}^i$
- The corresponding weight is $\mathbf{w}_k$
- We calculate the corresponding weight for every eigenface

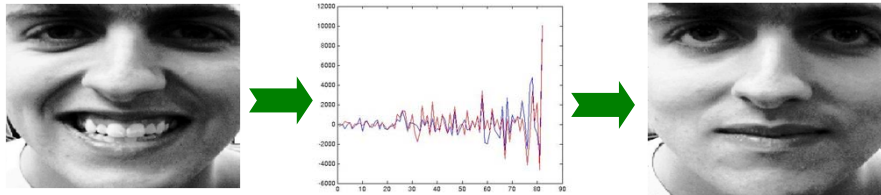


Recognition in face space

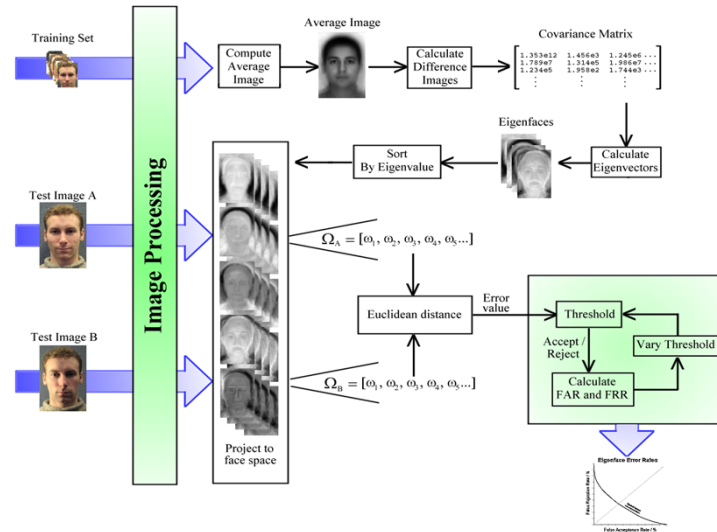
- Recognition is now simple. We find the euclidean distance d between our face and all the other stored faces in face space:

$$d(w^1, w^2) = \sqrt{\sum_{i=1}^n (w_i^1 - w_i^2)^2}$$

- The closest face in face space is the chosen match

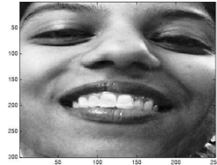


Test Procedure

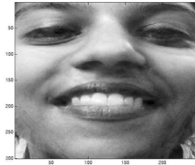


Reconstruction

- The more eigenfaces you have, the better the reconstruction, but you can have high quality reconstruction even with a small number of eigenfaces



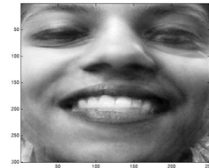
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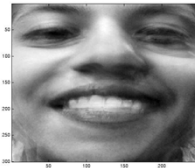
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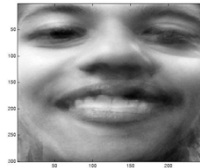
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Demo & Tutorial

<http://www.cs.princeton.edu/~cdecoro/eigenfaces/>

<http://www.pages.drexel.edu/~sis26/Eigenface%20Tutorial.htm#Programming>

Summary

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References

- J. Wyatt, “Face Recognition”, School of Computer Science, University of Birmingham.
- M. Turk and A. Pentland (1991). Eigenfaces for recognition, *Journal of Cognitive Neuroscience*, 3(1): 71–86.
- S. Theodoridis and K. Koutroumbas, *Pattern Recognition* (4th Edition), Academic Press, 2009.

